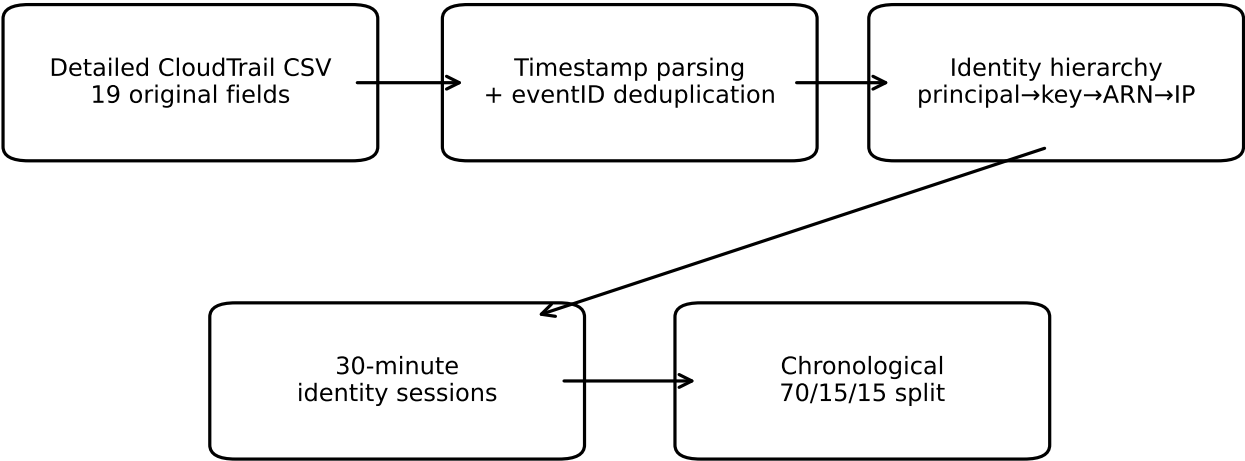
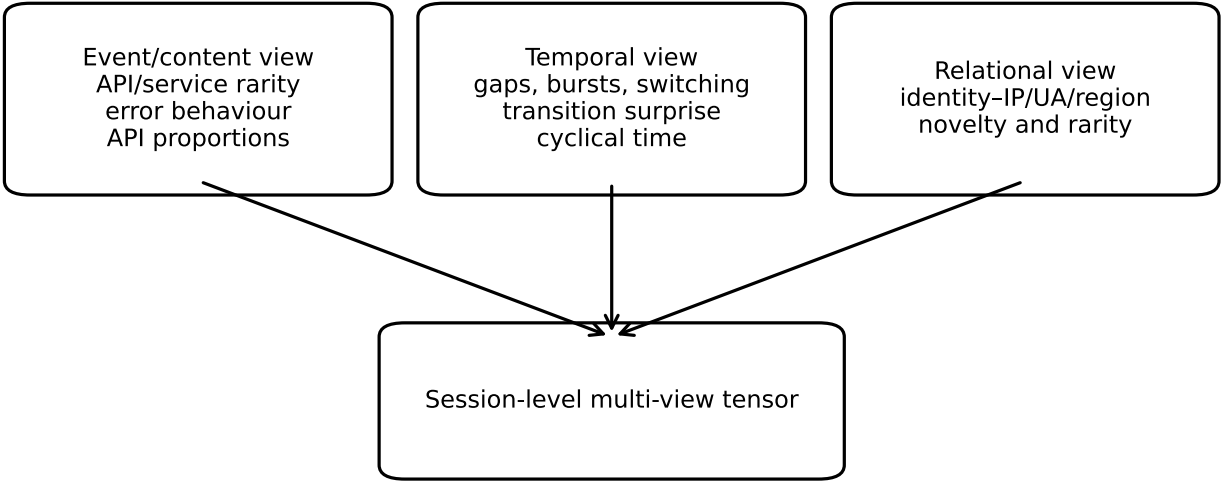


Figure 1. CloudTrace-MVAD data curation and evaluation workflow

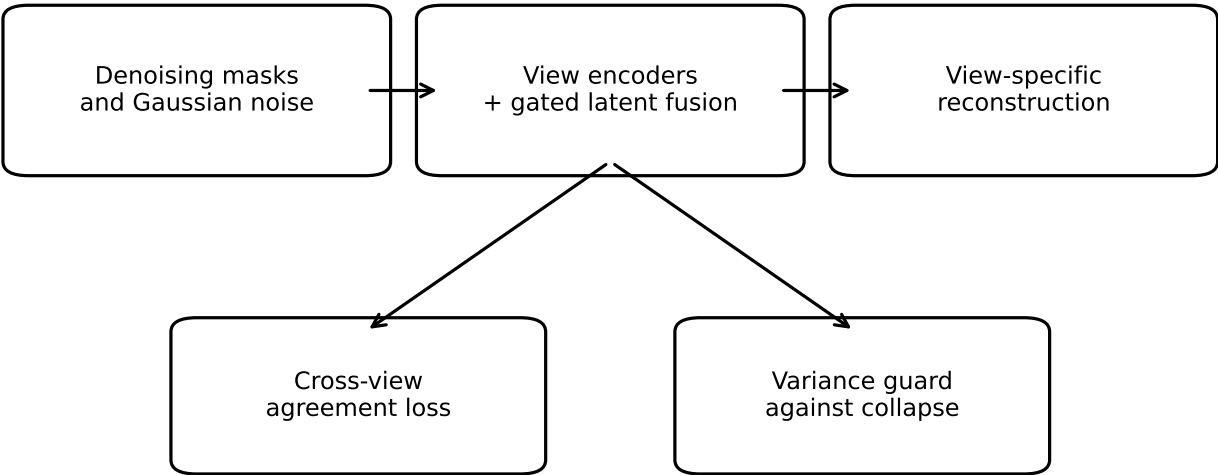
(a) Leakage-audited data curation



(b) Three complementary feature views



(c) Fully label-free learning



(d) Evaluation without human attack labels

